

opto-sim Research Roadmap

Physical-Layer Fiber-Optic Simulator

Core Principle

This paper models the physical components of an optical system sufficiently, and it aligns with published literature or actual physical components. Additionally these components are modular and can be attached with one another to produce relatively proper real-like graphs and parameters.

Every component is validated against the published formula or datasheet it implements — no more, no less. The paper demonstrates that the components compose correctly in a modular pipeline.

Publication Plan

#	Paper	Target	When
A	Software validation: component-wise validation + modular composition	OE/JLT/SoftwareX	Now
B	QKD environmental sensitivity	OE/JLT/EPJ QT	After A
C	Physical-layer security analysis	IEEE T-IFS/NPJ QI/PRX Q.	After B
D	Full QKD system (optional)	Nature Comms/PRX Q.	After C

Tier 0 — Infrastructure (DONE)

pytest suite, seeded reproducibility, 48 tests. No further work.

Tier 1 — Software Validation Paper

Paper claim: Each component matches its source literature, and they compose into a working end-to-end simulation.

1.1 Fiber Channel (DONE)

Validation	Reference	Error	Script
CD: Gaussian pulse broadening	Agrawal Fig. 2.6	0.0000%	validate_cd.py
PMD: DGD histogram vs Maxwellian	Razavi Fig. 2.11	KS $p = 0.31$	validate_pmd.py
Attenuation: $\alpha = 0.182 \text{ dB/km}$ @ 1550 nm	SMF-28 spec	0.0000%	validate_attenuation.py
Birefringence: $\Delta n \propto (r_f/R)^2$	Yuan Fig. 1	0.0000%	validate_birefringence.py

1.2 Laser (NEW — validation scripts needed)

Validation	Reference	What to check
RIN spectrum	Coldren Eq. 5.3.38	PSD peak frequency $\pm 10\%$, roll-off $1/f^2$
Phase noise linewidth	Henry Eq. 18	Fitted Lorentzian FWHM = input $\Delta\nu \pm 10\%$

No new physics. Models already implemented in `cwlaser.py`. Scripts confirm output matches the published formula.

1.3 MZM (NEW — validation script needed)

Validation	Reference	What to check
V_π from crystal geometry	Weis & Gaylord 1985	V_π within $\pm 5\%$ of published value
Extinction ratio	MZM model	On/off ratio matches user-specified ER

Already tested in `test_mzm.py`. One script for a publication figure.

1.4 APD (NEW — validation script needed)

Validation	Reference	What to check
Responsivity $R = \eta e \lambda / (hc)$	Kasap Eq. 4.19	R values match formula
Excess noise factor F on shot noise only	Kasap Eq. 4.45	$I_{\text{noise}}^2 = 2eIB \cdot F$, thermal not scaled

Already tested in `test_apd.py`. One script for a publication figure.

1.5 System Demonstration — Modular Composition (NEW)

Demo	What it proves
BB84 QBER vs distance with CD+PMD	All four components (laser, MZM, fiber, APD) working together
Link budget: laser power → fiber → detected power	Power flows correctly through pipeline

1.6 Effort Summary

What	Sessions	Status
Fiber validation (4 scripts)	—	✓
Laser validation (2 scripts)	~2	○
MZM / APD validation (2 scripts)	~1	○
System demo script	~1	○
Paper writing	—	○
Total	~4 sessions	

No new physics. No spectral models. No Sellmeier. No $D(\lambda)$ or $\alpha(\lambda)$ curves. Only validation scripts confirming each component matches its published source, and a demonstration that they work together.

Tier 2 — QKD Environmental Sensitivity (DEFERRED)

Requires Tier 1. Same scope as previous roadmap.

Tier 3 — Physical-Layer Security Analysis (DEFERRED)

Requires Tiers 1 and 2.

Tier 4 — Full QKD System Optimisation (OPTIONAL)

Requires Tiers 1-3.

Summary

Tier	What	Effort	Dependency		
0	Infrastructure	DONE	—		
1	Software validation	~4 sessions	—		
2	QKD environmental	6 w	Tier 1		
3	QKD security	5 m	Tiers 1-2		
4	Full QKD system	10 m	Tiers 1-3		

Advantage	opto-sim	NetSquid	qkdX	Comsis
Physical-layer field	✓	—	—	✓
Validated vs literature	<i>in prog.</i>	—	—	—
Modular composition	<i>in prog.</i>	—	—	—
Open-source (Python)	✓	✓	—	—
Laser RIN + phase noise	✓	—	—	—
Temp/bend birefringence	✓	—	—	—
FFT-based CD + PMD	✓	—	—	—
APD excess noise (Kasap)	✓	—	—	—
Seeded reproducibility	✓	—	—	—